

Technologies	Advantages	Disadvantages
RFID cards	Works automatically	Illicit usage and it is a problem when the student loses the RFID tag
Fingerprint scanners	Cannot be forgotten or lost	Time consuming for students to wait and give their attendance
Iris based	Very high accuracy	Expensive

Table 1: Advantages and disadvantages of various recognition technologies

always an issue as users can be easily tracked, and its coverage range is limited to only 3 metres.

Fingerprint attendance management system: The fingerprint system for attendance or for logging people's movements is widely used across the globe. This type of system has various advantages. Since humans' finger prints are unique, tracking them based on this is a highly secure and accurate system. But this type of system has many disadvantages too, such as the need for a fingerprint sensor to be placed in a location where there is proper wireless network connectivity. It is also much slower than many other approaches. Distortions caused by the glass layer on the sensor often lead to wrong results, which then require one to place one's finger repeatedly on the sensor to get proper results. This process is also very time consuming, as each student has to wait for his/her turn.

Iris based attendance system: This system uses technology that captures the image of the iris and checks to see if it matches with the data stored. Iris recognition is one of the most secure methods of recognition that can be used. It is easy, accurate, stable and has great scalability. It also offers the best privacy. But the difficulty occurs in laying transmission lines in places where the topography is challenging. This method also demands expensive hardware, which may not be viable for an attendance system. Since the iris is small, it is difficult to detect it from a certain distance. So the system needs to be set up in a place where people can

approach it easily. Besides, the constant use of this scanner might affect one's eyes as it uses infrared light to detect the iris in the dark. Sometimes, it gets difficult to scan the iris if there is too much reflection.

The different methods that can be used to set up an attendance system

Approach 1: In this approach, the student database, which contains images of students, is used as the training data for the face recognition algorithm.

A series of group images of the whole class is given as the input to the facial recognition algorithm, which outputs an Excel sheet with the names/roll numbers of the students present in the group photo. The drawback of this approach is that it is difficult to capture the group image of a class with a large strength.

Another major drawback is that all the students may not be looking at the camera while the image is captured and there is also a chance that a person sitting in a front row may cover the person sitting in the row behind. So, to overcome these drawbacks, we came up with a new approach.

Approach 2: In this approach, we used the student photo ID database as the training data. We captured the images of students individually while they were entering the classroom and stored them in a folder. We gave this folder as input to the facial recognition algorithm, which then responded with the result in the same way as in the previous example.

The major drawback of this approach is the inadequate memory to store the captured individual images of the students. Storing these images on an online database is also not feasible as searching and retrieving the data could be time consuming.

So we proposed another solution — a real-time facial recognition system for the attendance system which addresses the disadvantages mentioned in different ways.

Implementation

Step-by-step process for any facial recognition process: The major steps in any facial recognition process are to:

- *find* the face of the person in the image;
- *analyse* the facial features of the person;
- *compare* the analysed features of the persons with the known faces; and
- *predict* the name of the person based on the above comparisons.

This process uses a Python package named 'face recognition', which implements the 'history of oriented gradient' (HOG) algorithm. Figure 3 is the diagrammatic representation of the HOG feature extraction and face encodings.

An overview of how the HOG algorithm works is given below.

- To find the faces in the image, the algorithm starts by converting the image into monochrome because colour data is unnecessary for further processing.
- It considers each pixel at a time and checks the pixels that connect directly with (surround) the selected pixel.
- It figures out how dark the selected pixel is compared with pixels immediately surrounding it.
- Based on this, an arrow represents the direction in which the image is darker.
- After repeating this process for every pixel in the image, we end up getting an arrow in every pixel.